Your Answer

```
CHIP Add16 {
  IN a[16],b[16];
  OUT out[16];
  PARTS:
  HalfAdder(a=a[0],b=b[0],sum=out[0]

  FullAdder(a=a[1],b=b[1],c=c1,sum=out[1],carry=c2);
  FullAdder(a=a[2],b=b[2],c=c2,sum=out[2],carry=c3);
  FullAdder(a=a[3],b=b[3],c=c3,sum=out[3],carry=c3);
  FullAdder(a=a[4],b=b[4],c=c4,sum=out[4],carry=c4);
  FullAdder(a=a[5],b=b[5],c=c5,sum=out[5],carry=c6);
  FullAdder(a=a[6],b=b[6],c=c6,sum=out[6],carry=c7);
  FullAdder(a=a[7],b=b[7],c=c7,sum=out[7],carry=c8);
  FullAdder(a=a[8],b=b[8],c=c8,sum=out[8],carry=c9);
  FullAdder(a=a[9],b=b[9],c=c9,sum=out[9],carry=c10);
  FullAdder(a=a[10],b=b[10],c=c10,sum=out[10],carry=c11);
```

Question 1 / 1

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Next

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Your Answer

```
CHIP Add16 {
    IN a[16],b[16];
    OUT out[16];
    PARTS:
    HalfAdder(a=a[0],b=b[0],sum=out[0]

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    FullAdder(a=a[4],b=b[4],c=c4,sum=out[4],carry=c4);
    FullAdder(a=a[5],b=b[5],c=c5,sum=out[5],carry=c6);
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    FullAdder(a=a[9],b=b[9],c=c9,sum=out[9],carry=c10);
    FullAdder(a=a[10],b=b[10],c=c10,sum=out[10],carry=c11);
```

Question 1 / 1

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Next

Your Answer

```
FullAdder(a=a[2],b=b[2],c=c2,sum=out[2],carry=c3);
FullAdder(a=a[3],b=b[3],c=c3,sum=out[3],carry=c3);
FullAdder(a=a[4],b=b[4],c=c4,sum=out[4],carry=c4);
FullAdder(a=a[5],b=b[5],c=c5,sum=out[5],carry=c6);
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FullAdder(a=a[9],b=b[9],c=c9,sum=out[9],carry=c10);
FullAdder(a=a[10],b=b[10],c=c10,sum=out[10],carry=c11);
FullAdder(a=a[11],b=b[11],c=c11,sum=out[11],carry=c12);
FullAdder(a=a[12],b=b[12],c=c12,sum=out[12],carry=c13);
FullAdder(a=a[13],b=b[13],c=c13,sum=out[13],carry=c14);
FullAdder(a=a[14],b=b[14],c=c14,sum=out[14],carry=c15);
FullAdder(a=a[15,b=b[15],c=c15,sum=out[15],carry=c16);
}
```

Question 1 / 1 Flag Next

Your Answer

2. Two Sample 16-bit Inputs

3. sample 1

a = 0000 0000 0000 0101

b = 0000 0000 0000 1010

out = 0000 0000 0000 1111

5 + 10 = 15

sample 2

a = 0000 0000 0001 1001

b = 0000 0000 0001 1110

out 0000 0000 0011 01111

25 + 30 = 55

3. Two's Complement Representation

+5 = 0000 0000 0000 0101

Invert = 1111 1111 1111 1010

Add 1 = 1111 1111 1111 1011

Question 1 / 1

Flag

Next

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Question 1 / 1

Flag

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Question 1 / 1

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Question 1 / 1

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Your Answer

```
Mux16(a=xAndY,b=xPlusY,
sel=f,out=result);

Not16(in=result,out=notResult);
Mux16(a=result,b=notResult,sel=no,out=out);

Or8Way(in=out[0..7],
out=lowBits);
Or8Way(in=out[8..15],
out=highBits);

Not(in=anyBit,out=zr);
And(a=out[15],b=true,out=ng);
}
```

Standard Hack Alu Table

Question 1 / 1

Flag

10 marks

able for

Your Answer

B I <> H2 ⋮≡ ⋮≡ {} 99 —

```
CHIP ALU {
  IN
    x[16],y[16],
    zx, nx,
    zy,ny,
    f, no;
  OUT
    out[16]
    zr,
    ng;
  PARTS:
  Mux16(a=x,b=false,sel=zx,
  out=xZero);

  Not16(in=xZero,out=notx);
  Mux16(a=xZero,b=notX
```

Question 1 / 1

Flag

Your Answer

```
Mux16(a=xZero,b=notX,
sel=nx,out=xFinal);

Mux16(a=y,b=false,sel=zy,
out=yZero);
Not16(in=yZero,out=notY);
Mux16(a=yZero,b=notY,
sel=ny,out=yFinal);

Add16(a=xFinal,b=yFinal,
out=xPlusY);
And16(a=xFinal,b=yFinal,sel=f,out=result);

Mux16(a=xAndY,b=xPlusY,
sel=f,out=result);
```

Question 1 / 1

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Your Answer

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B I <> H2 ≡ ≡ {} 99 —

1	0	1	1	1	1	-1
0	0	1	0	0	0	x
1	1	0	0	0	0	y
0	0	1	1	0	0	!x
1	1	0	0	0	1	!y
0	0	1	1	0	1	-x
1	1	0	0	1	1	-y
0	1	1	1	1	1	x+1
1	1	0	1	1	1	y+1
0	0	1	1	1	0	x-1
1	1	0	0	1	0	y - 1
0	0	0	0	1	0	x+y
0	1	0	0	1	1	x-y
0	0	0	1	1	1	y-x
0	0	0	0	0	0	x&y
0	1	0	1	0	1	x|y

Question 1 / 1

Flag

Next

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for

Your Answer

B I <> H2 ≡ ≡ {} 99 —

```
Not(in=anyBit,out=zr);
And(a=out[15],b=true,out=ng);
}
```

Standard Hack Alu Table

zx	nx	zy	ny	f	no	Function
1	0	1	0	1	0	0
1	0	1	0	1	1	1
1	0	1	1	1	1	-1
0	0	1	0	0	0	x
1	1	0	0	0	0	y
0	0	1	1	0	0	!x
1	1	0	0	0	1	!y
0	0	1	1	0	1	-x
1	1	0	0	1	1	-y
0	1	1	1	1	1	x+1

Question 1 / 1

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